

Automated Train Detection and Gate Control System

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*A comprehensive Minor Project report has been submitted in partial fulfillment
of
the requirements for the degree of*

Bachelor of Technology

in

ELECTRONICS & COMMUNICATION ENGINEERING

Under the supervision of

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RCC INSTITUTE OF INFORMATION TECHNOLOGY

**Affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
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CERTIFICATE OF APPROVAL



This is to certify that the Minor Project titled “Automated Train Detection and Gate Control System” carried out by

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for the partial fulfillment of the requirements for B.Tech degree in **Electronics and Communication Engineering** from **Maulana Abul Kalam Azad University of Technology, West Bengal** is primarily based on their own work under the supervision of Dr. Apu Mistry. The contents of this Minor Project, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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DECLARATION



“We Do hereby declare that this submission is our own work conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute and that, to the best of our knowledge and belief, contains no material previously written by another neither person nor material (data, theoretical analysis, figures, and text) which has been accepted for the award of any other degree or diploma of the University or other Institute of higher learning, except where due acknowledgement has been made in the text.”

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CERTIFICATE of ACCEPTANCE



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<Signature of the External examiner with Date>

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INTRODUCTION

With the increasing dependence on railway transportation, ensuring safety at railway level crossings has become an important challenge. Railway level crossings are locations where railway tracks intersect with roads, making them highly vulnerable to accidents caused by delayed gate operation, human error, poor visibility, or the absence of an effective warning system. Although many railway crossings are automated today, several still rely on manual intervention or simple detection systems that are unable to handle complex operating conditions efficiently. Therefore, developing a reliable, intelligent, and cost-effective automated gate control system has become an important area of research.

The Automated Train Detection and Gate Control System presented in this project aims to improve the safety and efficiency of railway level crossings using an ESP32 microcontroller, vibration sensors, infrared (IR) sensors, servo motors, and buzzers. The proposed system automatically detects an approaching train, provides an audible warning to road users, closes the railway gates before the train reaches the crossing, and reopens them only after the train has completely cleared the crossing. This minimizes human intervention while improving operational reliability.

Unlike conventional student projects that are generally designed for a single railway track and a single direction of train movement, the proposed system is capable of supporting two closely spaced railway tracks with trains travelling in both clockwise and anticlockwise directions. The system determines the direction of the approaching train using strategically placed vibration sensors before initiating the gate control sequence. Based on the detected direction, the controller dynamically assigns different functions to selected IR sensors instead of dedicating separate sensors for each direction. This software-based sensor reuse reduces hardware requirements, minimizes GPIO usage, and simplifies the overall system design.

An important enhancement introduced in this project is the use of additional vibration sensors for reset validation. These sensors ensure that the system reset process can occur only after the train has reached the designated exit region. As a result, accidental triggering of the reset IR sensors due to environmental disturbances, human interference, or other unwanted objects cannot cause premature reopening of the gates. This additional validation significantly improves the reliability and safety of the system.

To further enhance safety, IR sensors are installed between the gates to detect any pedestrian or vehicle trapped inside the crossing area. If an obstruction is detected while the gates are closing or are already closed, the ESP32 immediately commands the servo motors to reopen the gates, allowing safe evacuation before normal operation resumes. This fail-safe mechanism ensures that automation never compromises human safety.

Overall, the proposed system demonstrates that intelligent software logic combined with optimized sensor placement can provide a reliable, low-cost, and scalable solution for railway gate automation. By supporting bidirectional operation, dual-track crossings, sensor validation, and obstruction detection while minimizing additional hardware, the project offers an effective prototype suitable for educational demonstration as well as future enhancement toward real-world railway automation.

LITERATURE REVIEW

Automation of railway level crossings has received considerable attention in recent years because of the need to reduce accidents caused by manual gate operation and delayed warnings. Various researchers have proposed automatic gate control systems using infrared sensors, ultrasonic sensors, vibration sensors, and microcontrollers to improve safety and reduce human intervention.

[P. Sai Kiran et al. \(2021\)](#) presented an "Automated Railway Gate Control System using Arduino and Ultrasonic Sensors." Their system uses two ultrasonic sensors placed before and after the crossing to detect the arrival and departure of a train. When the first sensor detects the train, the Arduino controller activates a servo motor to close the gate and sounds a buzzer to warn road users. Once the second sensor confirms that the train has completely crossed the level crossing, the gate is reopened automatically. The proposed system successfully automates gate operation but depends entirely on ultrasonic sensors and does not support bidirectional operation or multiple railway tracks.

[B. N. Kumar et al. \(2020\)](#) proposed a "Sensor Based Automatic Control of Railway Gate" that combines vibration sensors and infrared sensors for train detection. In this design, vibration sensors provide early detection of train movement by sensing vibrations in the railway track, while IR sensors accurately determine the train's position near the crossing. This combination improves reliability compared to systems using only optical sensors. However, the proposed system is primarily intended for a single-track railway crossing and assigns fixed functions to each sensor.

[M. S. Sanjay et al.](#) proposed an "Arduino-Based Automatic Railway Gate Control and Obstacle Detection System." Their work integrates infrared sensors for train detection with an ultrasonic sensor for obstacle detection between the gates. If a vehicle or pedestrian is detected within the crossing area, the controller prevents complete gate closure, thereby improving safety. This study highlights the importance of incorporating a fail-safe mechanism in automated railway crossings. However, the system does not address simultaneous train movements, bidirectional train operation, or hardware optimization through software-controlled sensor reuse.

The proposed Automated Train Detection and Gate Control System build upon the advantages of these existing systems while addressing their limitations. Instead of assigning permanent functions to individual sensors, the proposed system dynamically changes the role of selected IR sensors based on the direction of train movement. Furthermore, it supports bidirectional operation on two closely spaced tracks using a common level crossing without duplicating the entire sensor arrangement. Additional vibration sensors are introduced as reset-validation sensors, ensuring that the reset sequence can only occur after confirmed train movement, thereby preventing false resets caused by environmental disturbances or accidental sensor activation. These improvements provide a more reliable, scalable, and hardware-efficient solution compared with previously reported approaches.

METHOD

The project consists of several coordinated components working together for automated detection and gate control:

Hardware Architecture

- **SW420 Vibration Sensors:** Detects train-induced vibrations from a long distance.
- **IR Sensors:** Placed at intervals to detect the train's position sequentially.
- **ESP32:** Central processing unit to manage sensor inputs and output controls.
- **Servo Motor:** Physically opens or closes the gate.
- **Buzzer:** Provides audio signals indicating train proximity.
- **Internal Object Detection Sensor (IR Sensor):** Detects any object trapped in the gate path.

Working Flow

1. The ESP32 remains in idle mode while continuously monitoring all four vibration sensors.
2. V1 or V2 detects the approaching train and determines its direction of travel.
3. The corresponding IR sensor activates both buzzers to warn nearby vehicles and pedestrians.
4. The gate-control IR sensor signals the ESP32 to close both railway gates using the servo motors.
5. The remaining IR sensors monitor the train's movement through the crossing.
6. V3 or V4 validates the train's exit before allowing the system to reset.
7. IR5 and IR6 detect any obstruction between the gates and reopen them if necessary.
8. After the train clears the crossing, the gates reopen and the system returns to the idle state.

WORK DONE

The proposed Automated Train Detection and Gate Control System was built using an ESP32 microcontroller, vibration sensors, IR sensors, servo motors, and buzzers. The system was designed to automatically detect an approaching train, control the railway gates, and ensure safe operation under different conditions. Each component has a specific role in the working of the system, which is explained below.

Train Detection Using Vibration Sensors

The system uses four vibration sensors placed at different locations along the railway track. V1 and V2 are used to detect an approaching train and identify its direction of travel. The other two vibration sensors, V3 and V4, are used for reset validation. Before the system can reset after a train has crossed the gate, the corresponding vibration sensor must be triggered first as a fail-safe measure.

Train Detection Using IR Sensors

Six IR sensors are used to monitor the movement of the train around the railway crossing. The proposed system changes the function of some IR sensors depending on the direction of train movement. This reduces the number of sensors required while allowing the system to support trains moving in both directions. The remaining IR sensors are used for resetting the system and detecting obstacles between the railway gates.

ESP32 Microcontroller

The ESP32 is the main controller of the project. It continuously reads the inputs from all the vibration and IR sensors and controls the servo motors and buzzers according to the programmed logic. The ESP32 was chosen because it has sufficient GPIO pins, supports PWM for controlling servo motors, and offers higher processing speed than traditional Arduino boards. It also provides the flexibility to add wireless features in future if required.

Railway Gate Control

Two servo motors are used to operate the railway gates. When the train reaches the gate control sensor, the ESP32 sends PWM signals to both servo motors to close the gates automatically. The gates remain closed until the train has completely crossed the level crossing and all reset conditions have been satisfied. After that, the ESP32 commands both servo motors to reopen the gates.

Warning System

Two buzzers are used to warn vehicles and pedestrians about an approaching train. The buzzers are activated before the gates close and remain ON while the train passes through the crossing. Once the crossing becomes safe again, the buzzers are automatically switched OFF.

Obstacle Detection

Two IR sensors (IR5 and IR6) are placed between the railway gates to detect any vehicle or pedestrian trapped inside the crossing. If either sensor detects an obstruction while the gates are closing or already closed, the ESP32 immediately reopens the gates to allow safe passage. This feature improves the safety of the entire system.

Dual Track Operation

The proposed model supports two railway tracks sharing the same road crossing. The controller does not need to identify which track the train is travelling on because the objective is simply to prevent road traffic from entering the crossing whenever any train approaches. As long as a train is detected on either track, the gates remain closed until the crossing becomes completely clear.

Flow of Operation

- The system remains in the idle state and continuously monitors all vibration sensors.
- When V1 or V2 detects an approaching train, the ESP32 determines the direction of travel.
- The detected direction is locked until the current train completes its journey through the crossing.
- The corresponding IR sensor activates both buzzers to warn vehicles and pedestrians.
- The gate-control IR sensor commands the servo motors to close both railway gates.
- The train continues to move through the crossing while the ESP32 monitors the remaining IR sensors.
- V3 or V4 validates that the train has reached the exit section.
- The corresponding reset IR sensor resets the system only after successful validation.
- IR5 and IR6 monitor the crossing area for any obstruction.
- If an obstruction is detected, the gates are reopened immediately.
- After the train completely clears the crossing, the gates reopen and the system returns to the idle state.

Hardware Integration with ESP32

- The ESP32 is used as the main controller for processing all sensor inputs and controlling the output devices.
- Four vibration sensors are installed to detect train direction and validate the reset operation.
- Six IR sensors are placed around the crossing for train detection, gate control, reset, and obstacle detection.
- Two servo motors are used to automatically open and close the railway gates.

- Two buzzers provide an audible warning before gate closure.
- Sensors located on the same side of the track share common VCC and GND connections to reduce wiring complexity.
- The complete hardware is assembled on a compact prototype suitable for demonstration purposes.

Software Implementation

- The program is developed in C++ using the Arduino IDE.
- The ESP32 continuously reads data from all vibration and IR sensors.
- The software identifies the train direction using V1 and V2.
- A direction lock ensures that only one operating sequence is executed at a time.
- The functions of selected IR sensors change automatically depending on the detected direction.
- V3 and V4 validate the reset operation before the corresponding reset IR sensor is accepted.
- Servo motors and buzzers are controlled according to the programmed sequence.
- The software continuously checks IR5 and IR6 for obstacle detection and performs safety operations whenever required.

Testing Procedure

Testing of the proposed system was carried out using a scaled railway crossing prototype under different operating conditions. The following tests were performed:

- Direction Detection Test: Verified that V1 correctly detected clockwise movement and V2 correctly detected anticlockwise movement.
- Sensor Validation Test: Checked that the reset IR sensors (IR3 and IR4) could reset the system only after the corresponding validation vibration sensor (V3 or V4) had been triggered.
- Gate Operation Test: Repeatedly tested the opening and closing of both railway gates using the servo motors for trains travelling in both directions.
- Warning System Test: Verified that both buzzers were activated before gate closure and remained active until the train cleared the crossing.
- Obstacle Detection Test: Placed an object between the gates to ensure that IR5 and IR6 detected the obstruction and immediately reopened the gates.
- Dual Track Operation Test: Operated trains on both tracks to verify that the gates remained closed regardless of which track the train occupied.
- Continuous Operation Test: Performed multiple operating cycles to verify consistent performance, proper direction locking, and reliable system reset without false triggering.

HIGHLIGHTS

The proposed Automated Train Detection and Gate Control System incorporates several innovative features that improve the safety, reliability, and efficiency of railway level crossing automation while keeping the overall system simple and cost-effective.

Bidirectional Train Detection

Unlike conventional automatic railway gate systems that operate in only one direction, the proposed system supports trains travelling in both clockwise and anticlockwise directions. The ESP32 identifies the direction of the approaching train using vibration sensors and automatically follows the corresponding operating sequence without requiring any manual intervention.

Dual Track Operation

The prototype is designed to operate on two closely spaced railway tracks using a single gate control mechanism. Since the primary objective is to prevent road traffic from entering the crossing whenever a train approaches, the controller does not need to identify the specific track occupied by the train. This approach reduces hardware requirements while extending the functionality of the system.

Dynamic Sensor Assignment

Instead of assigning a permanent function to every IR sensor, the proposed system changes the function of selected sensors according to the detected direction of train movement. The same sensor can therefore perform different tasks during clockwise and anticlockwise operation. This significantly reduces the number of sensors required and makes better use of the available ESP32 GPIO pins.

Reset Validation Mechanism

One of the unique features of the proposed system is the use of **V3** and **V4** as reset validation sensors. Before the system is allowed to reset, the train must first activate the corresponding validation vibration sensor. This ensures that only the train which initiated the operating cycle can reset the system, preventing false resets caused by accidental IR sensor activation or external disturbances.

Obstacle Detection for Safety

The system includes two dedicated IR sensors (**IR5** and **IR6**) to monitor the area between the railway gates. If a vehicle, pedestrian, or any other object is detected while the gates are closing, the controller immediately reopens the gates. This feature provides an additional level of safety and reduces the risk of accidents at the railway crossing.

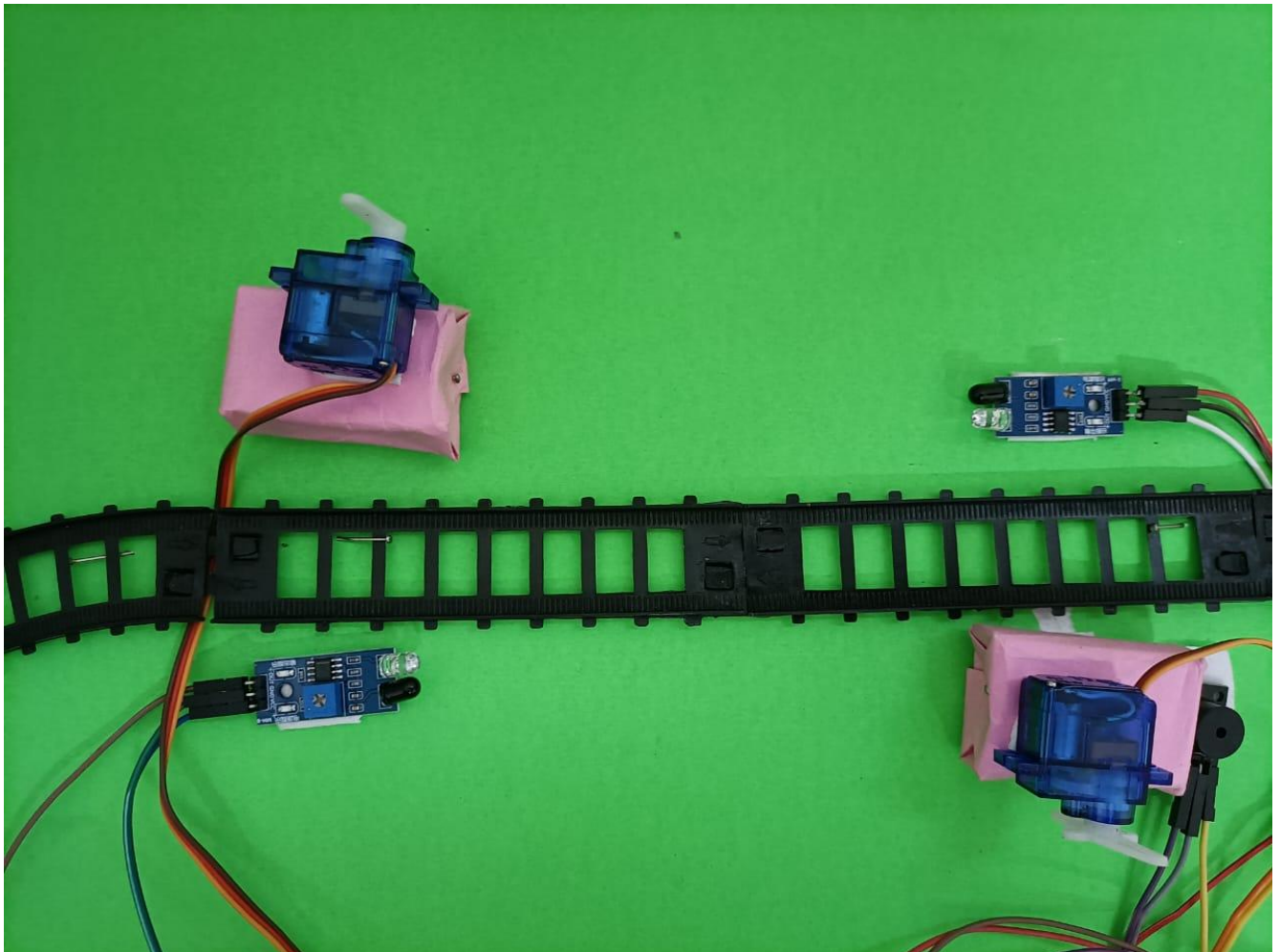
Reduced Hardware and Wiring

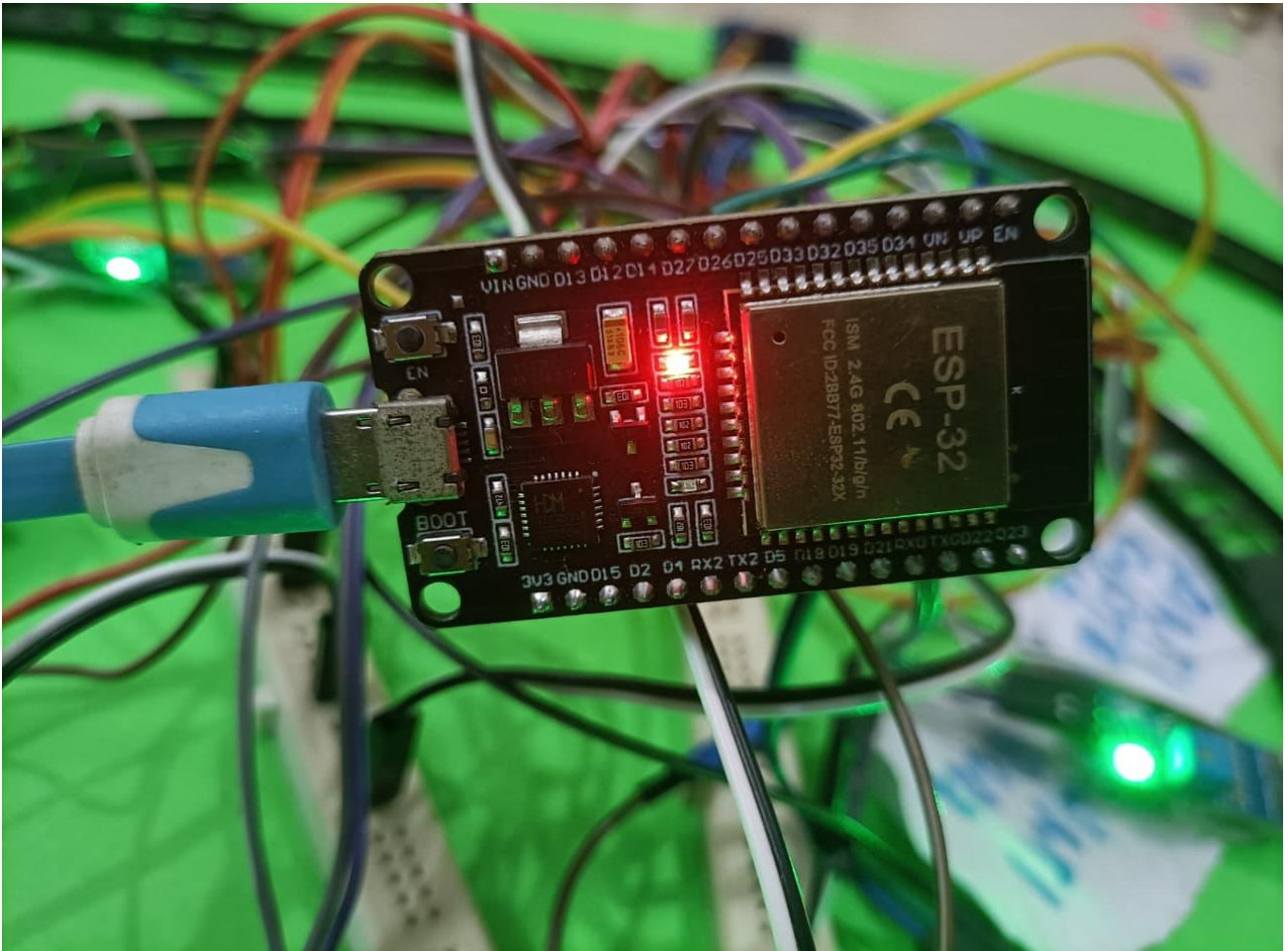
The project has been designed with hardware optimization in mind. Instead of increasing the number of sensors for every new feature, most improvements have been achieved through intelligent software logic. In addition, sensors located on the same side of the railway track share common VCC and GND connections, reducing the number of wires and making the prototype easier to assemble, transport, and maintain.

Reliable and Expandable Design

The use of the ESP32 microcontroller provides sufficient processing power and flexibility for future upgrades. Features such as GSM-based alerts, IoT monitoring, RFID train identification, cloud connectivity, or real-time railway monitoring can be added without major modifications to the existing system.

Visual model of the Internal Obstruction Safety Mechanism





ESP32

CONCLUSION

The Automated Train Detection and Gate Control System developed in this project successfully demonstrates a reliable and low-cost solution for automating railway level crossings using an ESP32 microcontroller. The system is capable of automatically detecting approaching trains, providing warning signals, operating the railway gates, and reopening them only after the train has safely cleared the crossing.

Unlike conventional student projects, the proposed system supports bidirectional train movement and dual-track operation while using a single gate control mechanism. By dynamically assigning different functions to the IR sensors based on the direction of train movement, the system reduces hardware requirements without affecting performance. The addition of reset validation sensors (V3 and V4) further improves reliability by ensuring that only the train which initiated the operating cycle can reset the system, thereby preventing false gate operations caused by accidental sensor activation.

The integration of obstacle detection using IR5 and IR6 provides an additional layer of safety by preventing the gates from closing when a vehicle or pedestrian is present inside the crossing. The use of shared VCC and GND connections also simplifies the wiring, making the prototype easier to assemble, maintain, and transport.

Overall, the project achieved all of its objectives and demonstrated stable operation during repeated testing. The proposed system is economical, easy to implement, and can serve as a foundation for future enhancements such as IoT-based monitoring, GSM alerts, RFID-based train identification, and integration with real railway signalling systems.

References:

1. [P. Sai Kiran *et al.*, "Automated Railway Gate Control System using Arduino and Ultrasonic Sensors," 2021.](#)
2. [B. N. Kumar *et al.*, "Sensor Based Automatic Control of Railway Gate," 2020.](#)
3. [M. S. Sanjay *et al.*, "Arduino Based Automatic Railway Gate Control and Obstacle Detection System."](#)